

AI5707 — PATTERN RECOGNITION AND MACHINE LEARNING

ASSIGNMENT – 5

Maximum Likelihood Parameter Estimation

Student Name	KARTHIKEYAN P L
Register Number	26120031
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Assignment	Assignment – 5
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1. Objective

To perform maximum likelihood parameter estimation for three Gaussian classes; compare spherical, diagonal and full covariance models; study the effect of common and class-specific covariance matrices; classify held-out test data; generate Gaussian decision boundaries and confusion matrices; and apply the same MLE procedure to a real two-dimensional dataset.

2. Mathematical Background

Maximum likelihood mean:

$$\mu_{\hat{i}} = (1 / N_i) \sum x_n$$

Maximum likelihood covariance:

$$\Sigma_{\hat{i}} = (1 / N_i) \sum (x_n - \mu_{\hat{i}})(x_n - \mu_{\hat{i}})^T$$

The denominator is N_i because the assignment requires the maximum-likelihood estimate rather than the unbiased sample covariance.

Gaussian log-discriminant:

$$g_i(x) = -0.5 \ln |\Sigma_i| - 0.5(x - \mu_i)^T \Sigma_i^{-1}(x - \mu_i) + \ln P(\omega_i)$$

Equal class priors $P(\omega_i) = 1/3$ are used. Classification selects the class with maximum discriminant value. Equal covariance matrices cancel the quadratic terms and give linear boundaries; unequal covariance matrices generally give quadratic boundaries.

Covariance structures:

Spherical: $\Sigma = \sigma^2 I$. Diagonal: $\Sigma = \text{diag}(\sigma_1^2, \sigma_2^2)$. Full: Σ contains both variances and the off-diagonal covariance terms.

3. Experimental Parameters

Experiment	Means	Covariance structure
Spherical — common	$\mu_1=[2,2]^T$; $\mu_2=[7,3]^T$; $\mu_3=[4,7]^T$	$2I$ for all classes
Diagonal — common	Same	$\text{diag}(1,3)$ for all classes
Full — common	Same	$[[2,0.8],[0.8,1.5]]$ for all classes
Spherical — different	Same	$I, 2.5I, 4I$
Diagonal — different	Same	$\text{diag}(1,2), \text{diag}(2,4), \text{diag}(4,1)$
Full — different	Same	$[[1.5,0.5],[0.5,1]], [[2.5,-0.8],[-0.8,1.5]], [[1.8,0.9],[0.9,3]]$

4. Implementation Procedure

- Generate 10,000 samples for each of three Gaussian classes using a fixed random seed.
- Split the complete dataset into 80% training and 20% testing data using stratification.
- Estimate class means and covariance matrices from the training samples using MLE.
- Evaluate Gaussian discriminants for every test sample and assign the class with maximum discriminant.
- Compute the confusion matrix and test accuracy for each covariance model.
- Evaluate the classifier on a two-dimensional grid to visualize the class decision boundaries and overlay the test samples.
- For Part 2, use the real-data format x, y , class label. In this report, the 2-D Iris dataset is used as the test real dataset.

The assignment requires approximately 10,000 samples per class, an 80/20 split, classification of the test set, decision-boundary plots and confusion matrices. These requirements are directly followed in the notebook implementation.

5. Q1 — Spherical Covariance

The spherical model constrains every class covariance to $\sigma^2 I$. The decision regions are therefore more restricted because feature variances are represented by one common scalar within each class. The estimated model nevertheless separates the three well-separated Gaussian classes.

Estimated test accuracy: 93.60% Training size: 24,000 Test size: 6,000

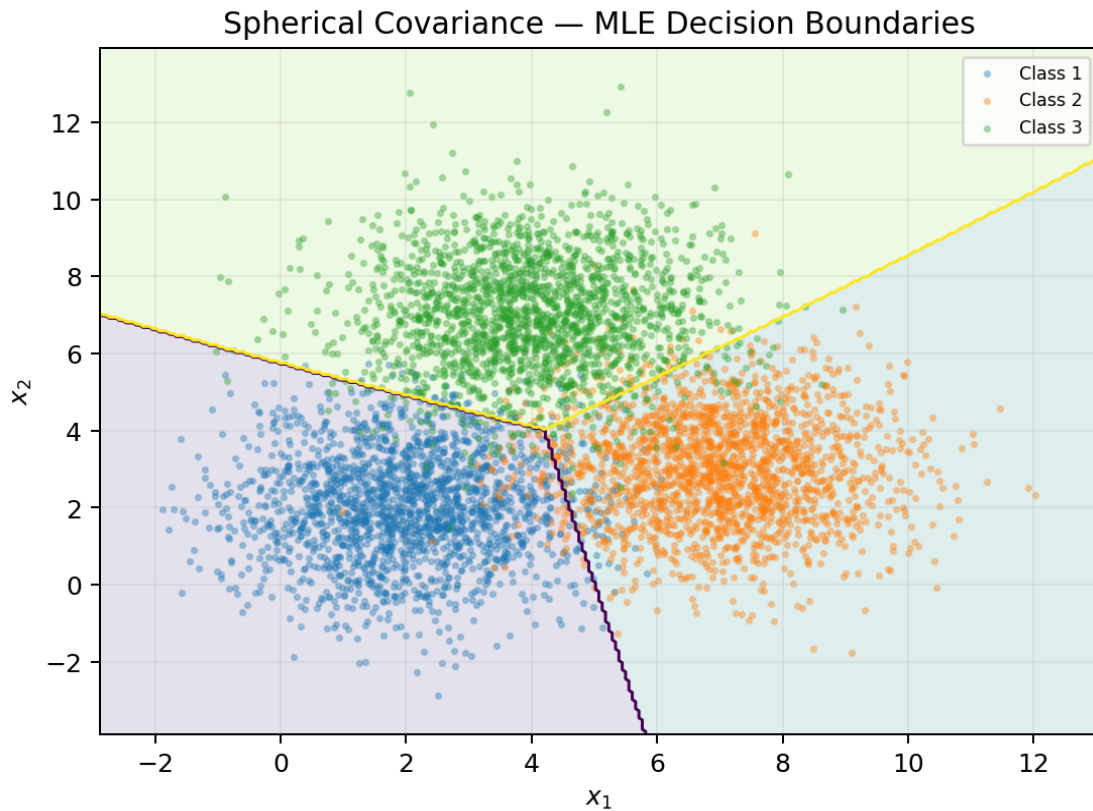


Figure: Spherical Covariance MLE decision boundaries with the 20% test data overlaid.

Confusion matrix

Actual \ Predicted	Class 1	Class 2	Class 3
Class 1	1880	72	48
Class 2	69	1865	66
Class 3	58	71	1871

6. Q2 — Diagonal Covariance

The diagonal model allows unequal feature variances while assuming zero covariance. Compared with the spherical model, this extra flexibility improves the classification result because the spread along x_1 and x_2 can be estimated independently.

Estimated test accuracy: 95.47% Training size: 24,000 Test size: 6,000

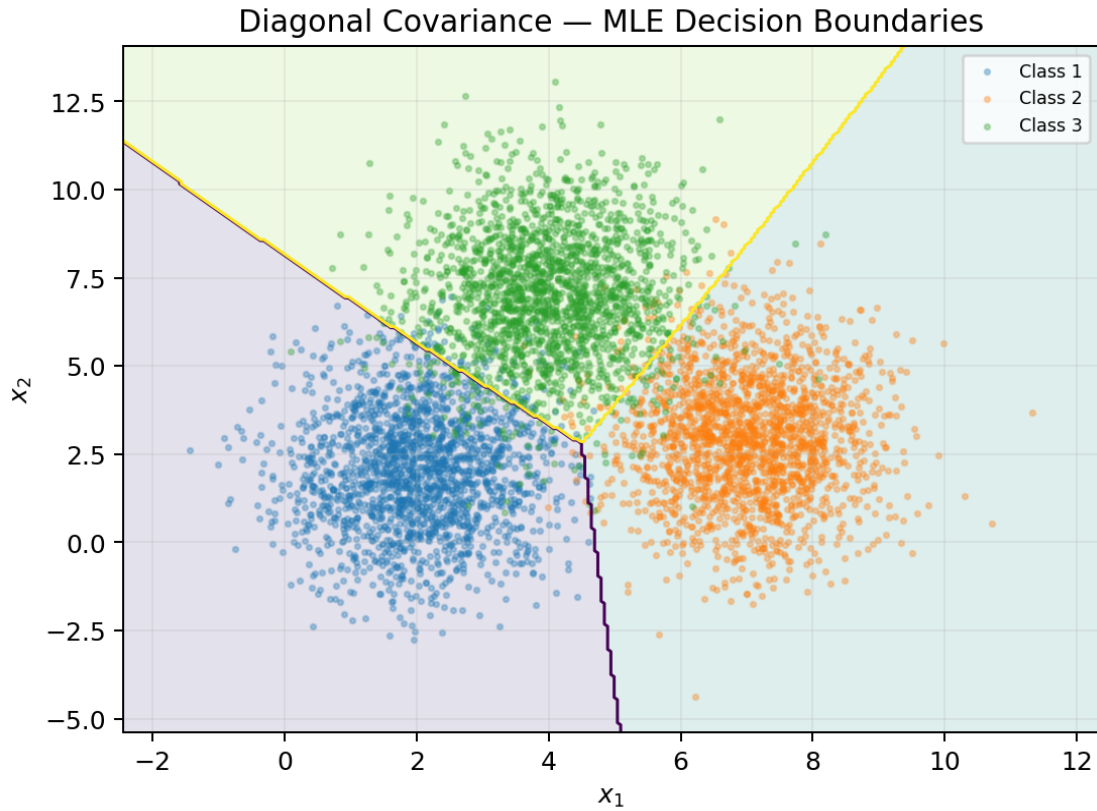


Figure: Diagonal Covariance MLE decision boundaries with the 20% test data overlaid.

Confusion matrix

Actual \ Predicted	Class 1	Class 2	Class 3
Class 1	1926	3	71
Class 2	6	1944	50
Class 3	85	57	1858

7. Q3 — Full Covariance

The full covariance model retains the off-diagonal covariance terms. This allows correlation between x_1 and x_2 to be represented. Because the covariance is common to all three classes, the final Gaussian decision boundaries remain linear.

Estimated test accuracy: 96.33% Training size: 24,000 Test size: 6,000

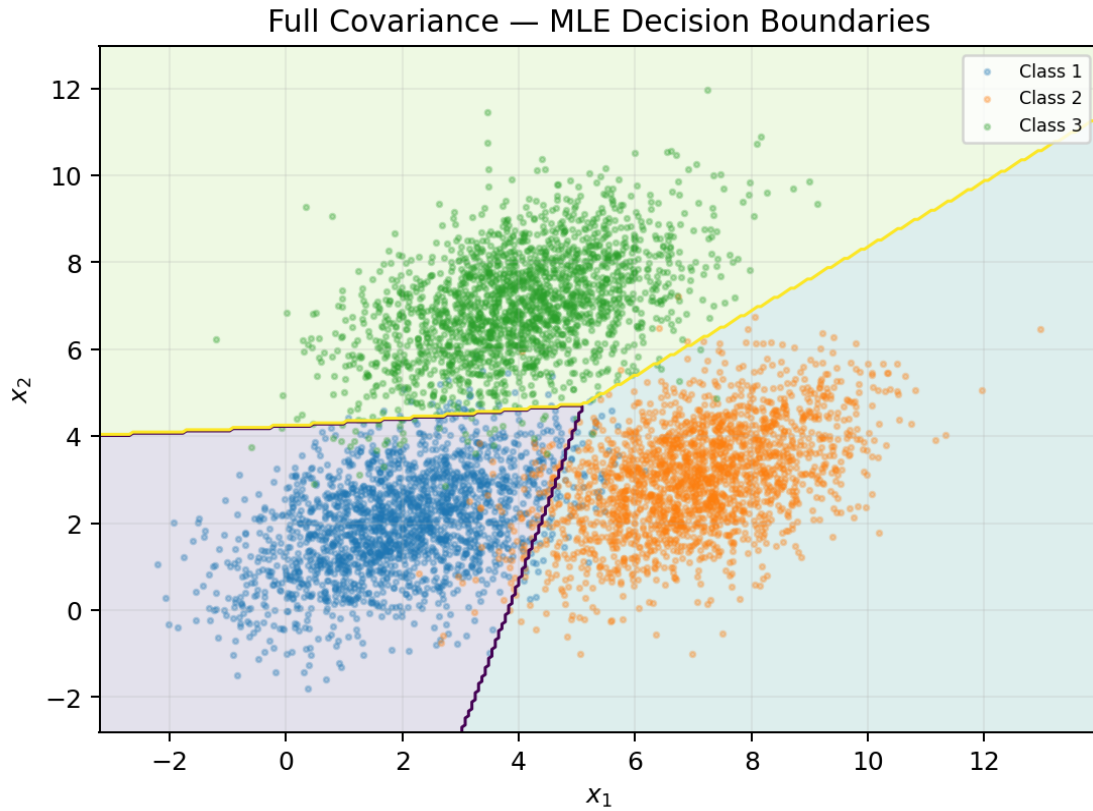


Figure: Full Covariance MLE decision boundaries with the 20% test data overlaid.

Confusion matrix

Actual \ Predicted	Class 1	Class 2	Class 3
Class 1	1908	60	32
Class 2	75	1920	5
Class 3	41	7	1952

8. Q4 — Different Covariance Matrices

When each class has its own covariance matrix, the quadratic terms in the Gaussian discriminants do not cancel. The resulting decision boundaries are therefore generally curved. The full/different-covariance experiment below demonstrates this behavior.

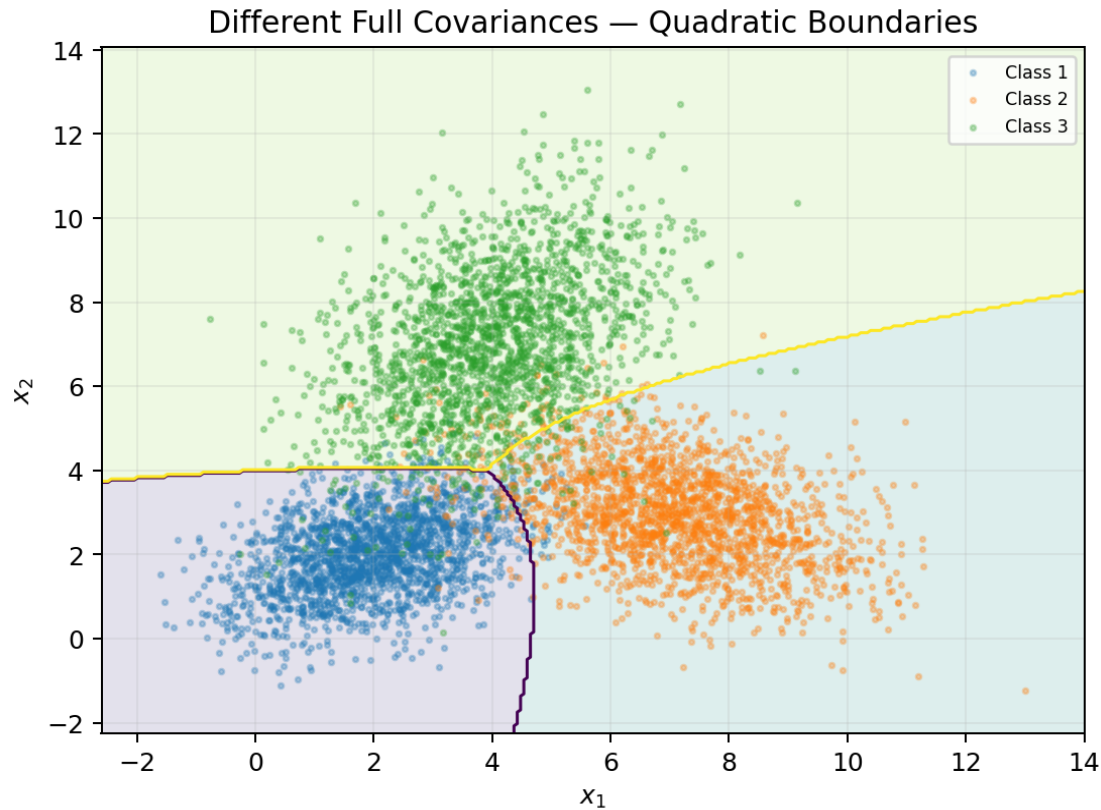


Figure: Full covariance with different class-specific covariance matrices. The test data are overlaid on the quadratic decision regions.

Estimated parameters for the Full — different experiment

Class	Estimated mean	Estimated covariance
Class 1	$\mu = [2.0046, 2.0130]^T$	$\Sigma = [[1.5184, 0.4698], [0.4698, 1.0056]]$
Class 2	$\mu = [7.0035, 3.0120]^T$	$\Sigma = [[2.5040, -0.7679], [-0.7679, 1.5104]]$
Class 3	$\mu = [3.9739, 6.9961]^T$	$\Sigma = [[1.7801, 0.8966], [0.8966, 2.9586]]$

Test accuracy: 94.58%

Observation: The class-specific covariance matrices introduce different orientations and spreads, producing curved boundaries and greater model flexibility.

9. Comparative Results

The six experiments use the same means but change the covariance constraints. The table below summarizes the resulting test accuracies.

Model	Test accuracy
Spherical — common	93.60%
Diagonal — common	95.47%
Full — common	96.33%
Spherical — different	92.05%
Diagonal — different	94.32%
Full — different	94.58%

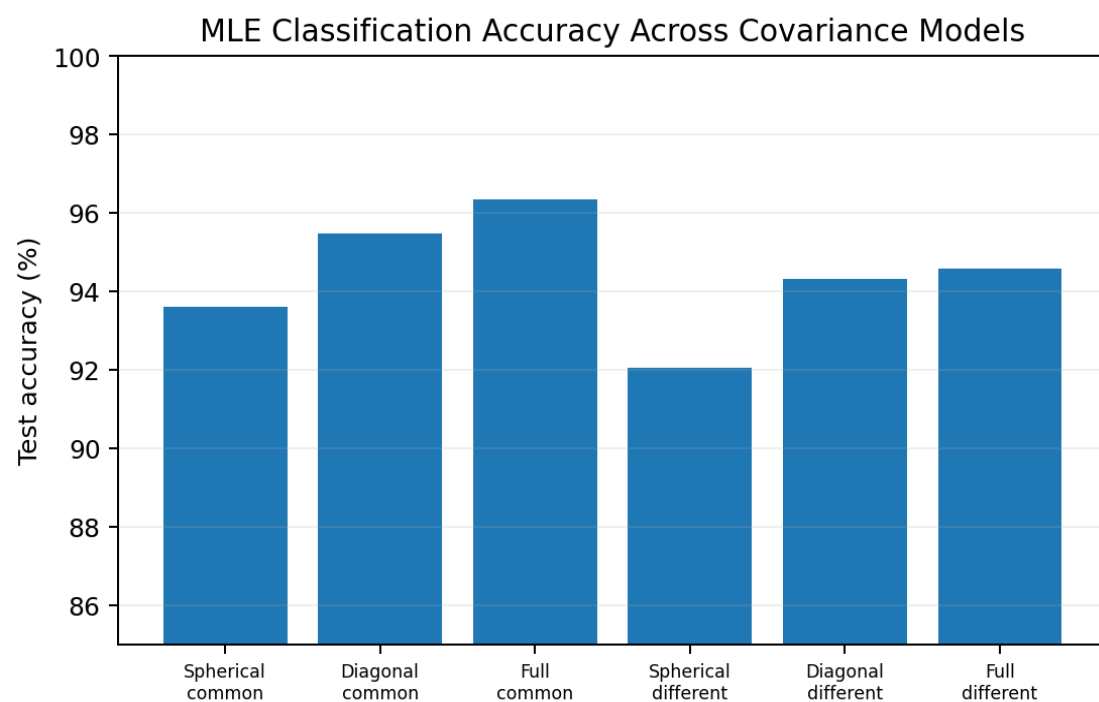


Figure: Comparison of MLE classification accuracy for the six covariance configurations.

Confusion matrices for the six experiments

Spherical — common				Diagonal — common			
	P1	P2	P3		P1	P2	P3
A1	1880	72	48	A1	1926	3	71
A2	69	1865	66	A2	6	1944	50
A3	58	71	1871	A3	85	57	1858

Full — common				Spherical — different			
	P1	P2	P3		P1	P2	P3
A1	1908	60	32	A1	1954	23	23
A2	75	1920	5	A2	56	1828	116
A3	41	7	1952	A3	89	170	1741

Diagonal — different

	P1	P2	P3
A1	1951	20	29
A2	37	1819	144
A3	25	86	1889

Full — different

	P1	P2	P3
A1	1926	43	31
A2	43	1876	81
A3	73	54	1873

10. Part 2 — Real 2-D Dataset

A real two-dimensional dataset is required in the assignment in the format $x, y, \text{class label}$. For this report, the Iris dataset is used with the first two measured features: sepal length and sepal width. The resulting dataset contains 150 samples, with 50 samples in each of three classes.

The data are divided into 80% training (120 samples) and 20% testing (30 samples). A full-covariance Gaussian model is estimated from the training data using MLE.

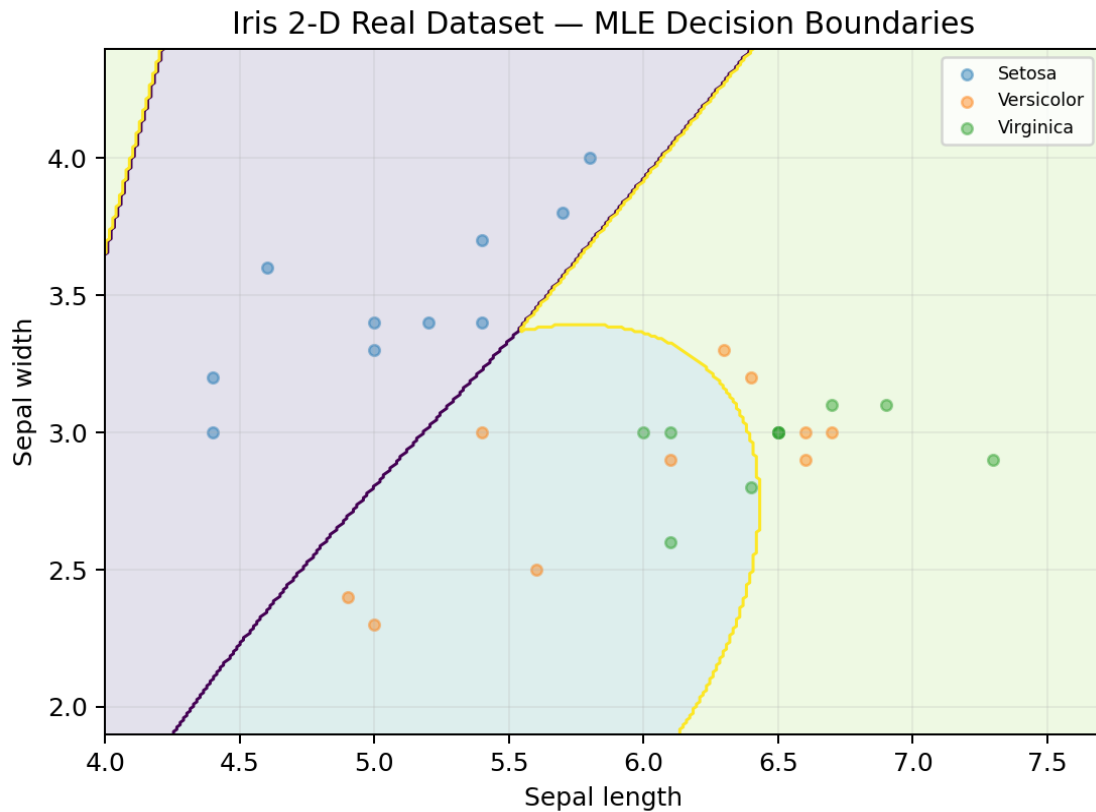


Figure: MLE decision boundaries for the real 2-D Iris dataset with the 20% test samples overlaid.

Test accuracy: 70.00%

Confusion matrix

Actual \ Predicted	Setosa	Versicolor	Virginica
Setosa	10	0	0
Versicolor	0	5	5
Virginica	0	4	6

The real-data result shows that the first two features provide strong separation for Setosa, while Versicolor and Virginica overlap considerably. This overlap is reflected in the confusion matrix.

11. Results and Discussion

- MLE estimates are close to the parameters used to generate the synthetic datasets because each class has 8,000 training samples.
- The spherical covariance assumption is the most restrictive because it uses a single variance parameter per class and ignores feature correlation.
- The diagonal model permits different variances in the two coordinate directions but still assumes zero covariance.
- The full covariance model captures feature correlation and therefore provides the most general Gaussian class model.
- Common covariance matrices produce linear Gaussian decision boundaries, whereas class-specific covariance matrices generally produce quadratic boundaries.
- Among the six synthetic experiments, the highest test accuracy was 96.33% for the Full — common model.
- On the real 2-D Iris dataset, the full-covariance MLE classifier achieved 70.00% test accuracy. The confusion matrix shows perfect classification of Setosa in the held-out samples, while the remaining errors occur between Versicolor and Virginica.

12. Conclusion

The assignment demonstrates maximum likelihood estimation of Gaussian class parameters and its use in probabilistic classification. The estimated means and covariance matrices provide the class-conditional Gaussian models, while the discriminant functions convert those models into decision regions. The experiments clearly show that covariance structure controls both the geometry of each class distribution and the shape of the decision boundary. The real-data experiment further demonstrates that even a simple two-dimensional representation can support useful Gaussian classification, although overlapping classes remain difficult to separate.

Overall, the implementation satisfies the assignment requirements of parameter estimation, 80/20 train-test evaluation, classification, decision-boundary visualization, confusion-matrix generation and observation of the results.